Poles and zeros lec-07

Convolution using z Transform:-

$x_1(n) \longleftrightarrow x_1(z)$

$x_2(n) \longleftrightarrow x_2(z)$

$x(n) = x_1(n) * x_2(n) \longleftrightarrow X(z) = x_1(z) \cdot x_2(z)$

Ex: 3.2.9 $x_1(n) = \{1, -2, 1\}$ (1 at $n=0$, 1 at $n=2$) $\quad x_2(n) = \begin{cases} 1 : 0 \le n \le 5 \\ 0 ; \text{elsewhere} \end{cases} = \{1, 1, 1, 1, 1, 1\}$

$$x_1(z) = \sum_{n=-\infty}^{\infty} x_1(n) z^{-n}$$

$$= \sum_{n=0}^{2} x_1(n) z^{-n} = x_1(0) z^{-0} + x_1(1) z^{-1} + x_2(2) z^{-2}$$

$$= 1 + (-2) z^{-1} + z^{-2} = 1 - 2z^{-1} + z^{-2}$$

$$X_2(z) = \sum_{n=0}^{5} x_2(n) z^{-n} = x_1(0) z^{-0} + x_1(1) z^{-1} + x_1(2) z^{-2} + x_1(3) z^{-3}$$

$$+ x_1(4) z^{-4} + x_1(5) z^{-5} + x_1(6) z^{-6}$$

$$= 1 + z^{-1} + 1z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6}$$

$$X(z) = x_1(z) \cdot x_2(z)$$

$$= (1 - 2z^{-1} + z^{-2})(1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6})$$

$$= 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6} - 2z^{-1} - 2z^{-2} - 2z^{-3} - 2z^{-4}$$

$$- 2z^{-5} - 2z^{-6} - 2z^{-7} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6} + z^{-7}$$

$$+ z^{-8}$$

$$= 1 + (-z^{-1}) + z^{-7} + z^{-8}$$ / Correct Ans: $1 - z^{-1} - z^{-6} + z^{-7}$

$$x(n) = \delta(n) - \delta(n-1) - \delta(n-6) + \delta(n-7)$$

$n = 0 \to 1$ $\quad n = +6 \to -1$

$n = 1 \to -1$ $\quad n = +7 \to 1$

$\{1, -1, 0, 0, 0, 0, -1, 1\}$ (1 at $n=0$, -1 at $n=1$, -1 at $n=6$, 1 at $n=7$)

Ans=

Rational z-Transform: A rational z-transform is a ratio of two polynomials in z (or z^{-1})

$$X(z) = \frac{N(z)}{D(z)}$$

Why rational z-Transform is important:-

(i) Analyze system (ii) Find stability (iii) Understand Frequency behaviour

(iv) Design Filter (FIR, IIR)

General Rational Form:

$$X(z) = \frac{N(z)}{D(z)} = \frac{b_0 + b_1 z^{-1} + \cdots + b_M z^{-M}}{a_0 + a_1 z^{-1} + \cdots + a_N z^{-N}} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

Factored Form: $H(z) = K \frac{(z-z_1)(z-z_2)\cdots(z-z_M)}{(z-p_1)(z-p_2)\cdots(z-p_N)}$

Poles & zeros

$$X(z) = \frac{N(z) \rightarrow \text{zero } N(z)=0}{D(z) \rightarrow \text{Poles } D(z)=0}$$

(i) Near zero → output becomes small

(ii) Near pole → output becomes very large

(iii) zero → Signal killers

(iv) pole → signal boosters

Det poles & zeros

Ex: 3.3.1 → $X(n) = a^n u(n);\ a > 0$

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = a^0 z^{-0} + a^1 z^{-1} + a^2 z^{-2} + \cdots$$

$$= 1 + az^{-1} + a^2 z^{-2} + \cdots = \frac{1}{1-az^{-1}} = \frac{z}{z-a}$$

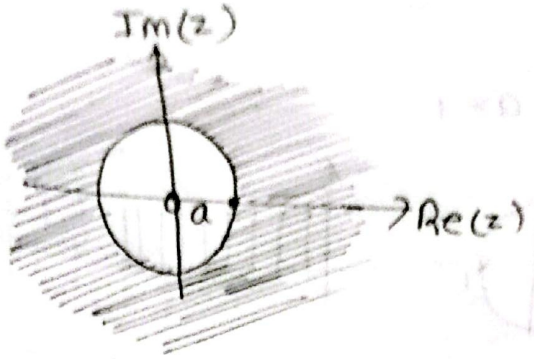
ROC: $|z| > |a|$

Zeros $z = 0$

Poles $z - a = 0$

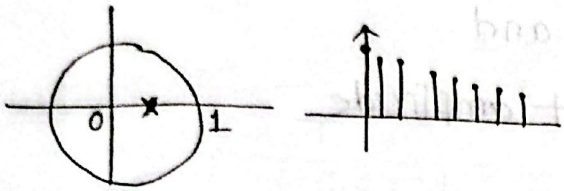
$\Rightarrow z = a$

Ans



Pole location and behaviour

Case 1 Pole located $0 < a < 1$

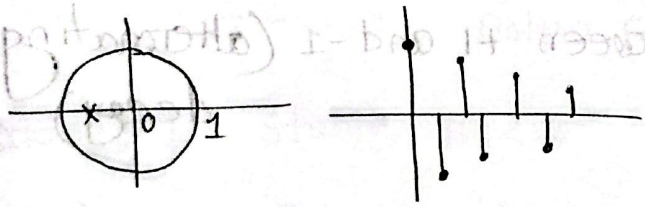


(i) Inside unit circle (stable)

Equation $x(n) = a^n$ (decays towards 0) $|p| < 1$

(iii) Time response is monotonically decreasing (non oscillation)

Case: 2 Pole at $a = -\pi$ & $|a| < 1$

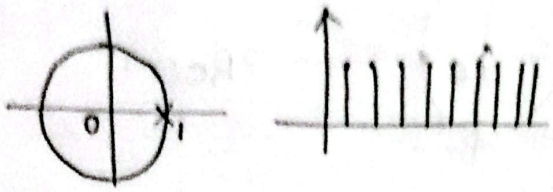


(i) Negative pole inside unit circle

$x(n) = -\pi^n$ (stable)

(ii) Signal decays with alternating sign (positive-negative oscillation)

Case: III $a = 1$

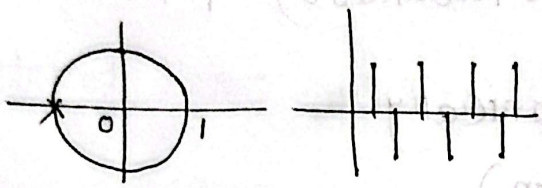


(i) Pole on unit circle positive

$x(n) = 1^n = 1$ (constant signal)

(ii) signal is non-decaying and non-growing → constant amplitude over the time

Case: IV $a = -1$

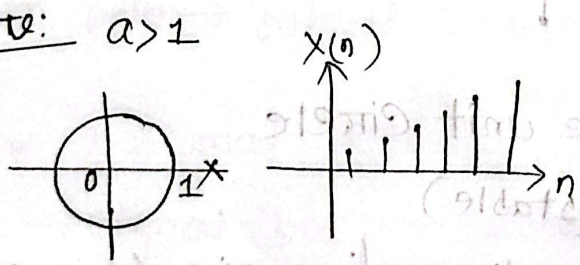


(i) Pole on unit circle negative

$x(n) = (-1)^n$

(ii) signal oscillates between +1 and -1 (alternating, no decay)

Case: v: $a > 1$

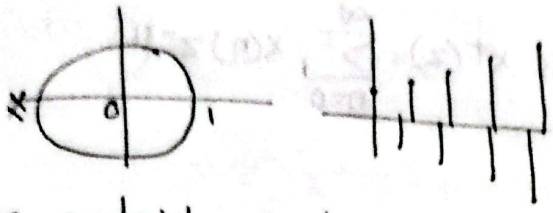


(i) Pole outside unit circle, positive.

$x(n) = a^n$ (exponentially grows)

(ii) Signal is monotonically increasing (no sign change, unstable growth).

Case: VI $a = -r$, $r > 1$



① pole outside unit circle, negative

$x(n) = (-r)^n$

② Signal grows with alternating signs (oscillatory, unstable).

Practice pb1

$$H(z) = \frac{1}{1+0.8z^{-1}} = \frac{z}{z+0.8}$$

Zeros $z = 0$ Poles $z + 0.8 = 0$ or $z = -0.8$

Stability: Stable ($|z = 0.8| < 1$)

Behavior: Decaying oscillator

Practice pb2

$$H(z) = \frac{z+0.9}{z^2 - 1.8z + 0.81} = \frac{z+0.9}{(z-0.9)(z-0.9)}$$

Zeros: $z = -0.9$ Poles $p = 0.9$

Stability: Stable $|p| < 1$

Behavior: Decreasing, non-oscillatory

Practice pb3

$$H(z) = \frac{z+0.5}{(z-0.6)(z-1.3)}$$

Zeros: $z = -0.5$ Poles $p = +0.6, +1.3$

Stabile: Unstable ($|p| > 1$)

Behavior: Growing (dominated by 1.3^n)

One sided z Transform lec-08

One sided z-transform $x^+(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$

Used for:

(i) Causal system
(ii) Initial value Problem
(iii) Difference equation
(iv) Discrete time control system

Ex: 3.5.1 $x_1(n) = \{1, 2, 5, 7, 0, 1\}$

$$x_1^+(z) = \sum_{n=0}^{5} x_1(n) z^{-n}$$

$$= x_1(0) z^{-0} + x_1(1) z^{-1} + x_1(2) z^{-2} + x_1(3) z^{-3} + x_1(4) z^{-4} + x_1(5) z^{-5} + x_1(6) z^{-6}$$

$$= 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$$ Ans

(ii) $x_2(n) = \{1, 2, \underset{\uparrow}{5}, 7, 0, 1\}$

$$x_2^+(z) = \sum_{n=0}^{3} x_2(n) z^{-n} = x_2(0) z^{-0} + x_2(1) z^{-1} + x_2(2) z^{-2} + x_2(3) z^{-3}$$

$$= 1 + 2z^{-1} + 5 + 7z^{-1} + z^{-3}$$ Ans

Time shifting Property:

$$z^+\{x(n-k)\} = z^{-k} x^+(z) + \sum_{m=1}^{k} x(-m) z^{m-k}$$

If causal

$$z^+\{x(n-k)\} = z^{-k} x^+(z)$$

Again $z^+\{x(n+k)\} = z^k x^+(z) - \sum_{m=0}^{k-1} x(m) z^{k-m}$

One sided z Transform lec-08

One sided z-transform $x^+(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$

Used for:

① Causal system
② Initial value Problem
③ Difference equation
④ Discrete time Control system

Ex: 3.5.1 $x_1(n) = \{1, 2, 5, 7, 0, 1\}$

$$x_1^+(z) = \sum_{n=0}^{5} x_1(n) z^{-n}$$

$$= x_1(0) z^{-0} + x_1(1) z^{-1} + x_1(2) z^{-2} + x_1(3) z^{-3} + x_1(4) z^{-4} + x_1(5) z^{-5} + x_1(6) z^{-6}$$

$$= 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$$

Ans

Ⓑ $x_2(n) = \{1, 2, 5, 7, 0, 1\}$ (↑ at 5)

$$x_2^+(z) = \sum_{n=0}^{3} x_2(n) z^{-n} = x_2(0) z^{-0} + x_2(1) z^{-1} + x_2(2) z^{-2} + x_2(3) z^{-3}$$

$$= 1 + 2z^{-1} + 5 + 7z^{-1} + z^{-3}$$

Ans

Time shifting Property:

$$z^+\{x(n-K)\} = z^{-K} x^+(z) + \sum_{m=1}^{K} x(-m) z^{m-K}$$

If causal

$$z^+\{x(n-k)\} = z^{-K} x^+(z)$$

Again $z^+\{x(n+K)\} = z^K x^+(z) - \sum_{m=0}^{K-1} x(m) z^{K-m}$

Pb 1 $x(n) = \left(\frac{1}{2}\right)^n u(n)$

$$z^+\{x(n-2)\} = z^{-2}X^+(z) + \sum_{m=1}^{K} x(-m) z^{m-K}$$

$$= z^{-2}X^+(z) + x(-1) z^{1-K} + x(-2) z^{2-K}$$

$$= z^{-2}X^+(z) + x(-1) z^{1-2} + x(-2) z^{2-2}$$

$$= z^{-2}X^+(z) + x(-1) z^{-1} + x(-2) \quad \text{---(1)}$$

$$X^+(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n . z^{-n} = \frac{1}{1 - 0.5 z^{-1}} = \frac{z}{z - 0.5} \quad \text{ROC } |z| > \left|\frac{1}{2}\right|$$

Putting the value of $X^+(z)$ in equation (1) and in causal system $x(-1)$ and $x(-2) = 0$

$$z^+\{x(n-2)\} = z^{-2} \frac{z}{z-0.5} = \frac{z^{-2+1}}{z-0.5} = \frac{z^{-1}}{z-0.5} \quad \text{Ans}$$

Ex: 3.5.2 (a) $x(n) = a^n u(n)$ $X^+(z) = ?$

$$X^+(z) = \sum_{n=0}^{\infty} x(n) z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1 - a z^{-1}} \quad \text{ROC: } |z| > |a|$$

Ans

(b) $x_1(n) = x(n-2)$ where $x(n) = a^n$

$$z^+\{x(n-2)\} = z^{-2}X^+(z) + x(-1) z^{1-2} + x(-2) z^{2-2}$$

$$= z^{-2} \frac{z}{z-a} + a^{-1} z^{-1} + a^{-2} z^0$$

$$= \frac{z^{-1}}{z-a} + a^{-1} z^{-1} + a^{-2}$$

Ans

Let,
$x(n) = a^n$
$x(-1) = a^{-1}$
$x(-2) = a^{-2}$

Practice pb2

$$z^+\{x(n+k)\} = z^k X^+(z) - \sum_{m=0}^{k-1} x(m) z^{k-m}$$

$$x(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$X^+(z) = \frac{1}{1-0.5z^{-1}} = \frac{z}{z-0.5} \quad \text{ROC: } |z| > |0.5|$$

$$z^+\{x(n+1)\} = z^1 \frac{z}{z-0.5} - x(0) z^1 - x(1) z^0$$

$$= \frac{z^2}{z-0.5} - z = \frac{z^2 - z^2 + 0.5z}{z-0.5} = \frac{0.5z}{z-0.5}$$

Ans

Practice Problem 03

Use one sided z-transform to solve the following difference equation with the given initial condition: $y(n+1) - 0.5y(n) = x(n)$; $n \geq 0$

Given: $y(0) = 2$

Determine the output transform $Y(z)$ in terms of the input transform of $X(z)$.

$$x(n) = y(n+1) - 0.5y(n)$$

$$X^+(z) = z^1 y^+(z) - y(0) z^1 - 0.5 y^+(z)$$

$$X^+(z) = z y^+(z) - 2z - 0.5 y^+(z)$$

$$X^+(z) + 2z = y^+(z)(z - 0.5)$$

$$y^+(z) = \frac{X^+(z) + 2z}{z - 0.5}$$

Ans

Pb4 $x(n) = (0.6)^n u(n)$

(i) $z^+\{x(n-1)\}$

$$X^+(z) = \frac{1}{1-0.6z^{-1}} = \frac{z}{z-0.6} \quad \text{ROC: } |z| > |0.6|$$

$$z^+\{x(n-k)\} = z^{-k} X^+(z) + \sum_{m=1}^{k} x(-m) z^{m-k}$$

$$= z^{-k} X^+(z) + x(-1) z^{1-k}$$

$$z^+\{x(n-1)\} = z^{-1} \frac{z}{z-0.6} + x(-1) z^{1-1}$$

$$= \frac{1}{z-0.6} + 0 = \frac{1}{z-0.6} \quad \text{Ans}$$

(ii) $z^+\{x(n+1)\} = z^1 X^+(z) - x(0) z^{1-0}$

$$= z \frac{z}{z-0.6} - z$$

$$= \frac{z^2}{z-0.6} - z = \frac{z^2 - z^2 + 0.6z}{z-0.6}$$

$$= \frac{0.6z}{z-0.6} \quad \text{Ans}$$

Pb5 $y(n) - 0.5y(n-1) = u(n) \quad ; \; y(-1) = 1$

$$\Rightarrow y^+(z) - 0.5\{z^{-1} y^+(z) + y(-1) z^{1-1}\} = \frac{1}{1-z^{-1}}$$

$$\Rightarrow y^+(z) - 0.5 z^{-1} y^+(z) - 0.5(1) z^0 = \frac{1}{1-z^{-1}}$$

$$\Rightarrow y^+(z) - 0.5 z^{-1} y^+(z) - 0.5 = \frac{1}{1-z^{-1}}$$

$$\Rightarrow y^+(z)\{1 - 0.5z^{-1}\} = \frac{1}{1-z^{-1}} + 0.5 = \frac{z}{z-1} + 0.5$$

$$= \frac{z + 0.5z - 0.5}{z-1}$$

$$\Rightarrow y^+(z)\{1-0.5z^{-1}\} = \frac{0.6z-0.5}{z-1}$$

$$\Rightarrow y^+(z) = \frac{0.6z-0.5}{(z-1)(1-0.5z^{-1})}$$

By taking partial fraction: -

$$\frac{A\,0.6z-0.5}{(z-1)(1-0.5z^{-1})} = \frac{A}{z-1} + \frac{B}{(1-0.5z^{-1})}$$

$$\Rightarrow 0.6z-0.5 = A(1-0.5z^{-1}) + B(z-1)$$

$$\Rightarrow 0.6z-0.5 = A-0.5Az^{-1}+Bz-B$$

$$\Rightarrow 0.6z-0.5 = (A-B)-0.5Az^{-1}+Bz$$

By taking coefficient of z, z^{-1} and constant

$B = 0.6$ $\quad A-B = -0.5$

$A = B-0.5 = 0.6-0.5 = 0.1$

$$\Rightarrow y^+(z)\{1-0.5z^{-1}\} = \frac{1+0.5-0.5z^{-1}}{1-z^{-1}}$$

$$\Rightarrow y^+(z) = \frac{1.5-0.5z^{-1}}{(1-z^{-1})(1-0.5z^{-1})}$$

By taking partial fraction: -

$$\frac{1.5-0.5z^{-1}}{(1-z^{-1})(1-0.5z^{-1})} = \frac{A}{1-z^{-1}} + \frac{B}{1-0.5z^{-1}}$$

$$\Rightarrow (1.5-0.5z^{-1}) = A(1-0.5z^{-1}) + B(1-z^{-1})$$

$$\Rightarrow 1.5-0.5z^{-1} = A-0.5Az^{-1}+B-Bz^{-1}$$

$$\Rightarrow 1.5-0.5z^{-1} = (A+B)-z^{-1}(0.5A+B)$$

$A+B = 1.5$ $\quad -0.5 = -(0.5A+B)$

$A = 1.5-B$ $\quad 0.5 = -0.5(1.5-B)-B$

$0.5 = -0.75+0.5B-B$

$\Rightarrow 0.5+0.75 = 0.4$

$\Rightarrow B = -0.5$

$A = 2$

Calculation গুলকা ঠিক আছে Process Correct

$$y+(z) = \frac{2}{1-z^{-1}} * \frac{0.5}{1-0.5z^{-1}}$$

$\therefore$ Inverse z-transform $z^{-1}\left\{\frac{1}{1-z^{-1}}\right\} = u(n)$

$$z^{-1}\left\{\frac{1}{1-0.5z^{-1}}\right\} = (0.5)^n u(n)$$

$$y(n) = 2u(n) - 0.5(0.5)^n u(n)$$

Ans

Lec-9

Q Why do we need Fourier analysis: -

Ans: In signal processing, many signal look complicated in time domain but become much simpler when we study them in terms of sinusoids or complex exponential.

Periodic signal: A signal $x(t)$ is periodic if there exist a positive no of T such that

$$x(t+T) = x(t) \text{ of all } t.$$

T = fundamental period = time taken for one full ~~repetition~~ repetition.

f_0 = fundamental frequency

$= \frac{1}{T}$ = cycles per second.

$\omega_0 = \frac{2\pi}{T}$ = fundamental angular frequency

= radians per second.

Orthogonality:

The inner product becomes zero.

Over T period $\int_T \cos(m\omega_0 t)\cos(n\omega_0 t)\,dt = 0$ if $m \neq n$

similarly for sine & cosine formula

This is very important because orthogonality allows us to "extract" each frequency component independently.

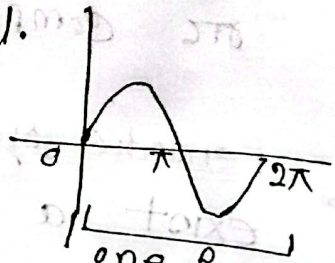
Euler's Formula:

$$e^{j\theta} = \cos\theta + j\sin\theta \qquad \cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$$

$$e^{-j\theta} = \cos\theta + j\sin\theta \qquad \sin\theta = \frac{e^{j\theta}+e^{-j\theta}}{2j}$$

Fourier series:

F.S applied to the periodic signal.



Trigonometric FS:-

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + b_n \sin n\omega_0 t$$

where, $\omega_0 = \frac{2\pi}{T}$

$a_0 = \frac{1}{T}\int_T x(t)\,dt$ = Avg or DC value of the signal

$a_n = \frac{2}{T}\int_T x(t)\cos n\omega_0 t\,dt$ = amount of cosine at frequency $n\omega_0$

$b_n = \frac{2}{T}\int_T x(t)\sin n\omega_0 t\,dt$

= amount of sin at frequency $n\omega_0$

Harmonics They are multiples the fundamental frequency ω_0, $2\omega_0$, $3\omega_0$, ...

Practice Problem 1

$$x(t) = 2\cos(\omega_0 t) + \frac{1}{2}\cos(3\omega_0 t) + \frac{1}{4}\cos(5\omega_0 t)$$

1. Strongest Harmonics = $2\cos(\omega_0 t)$ due to the highest amplitude

2. Higher harmonics creates faster oscillations and sharper changes in the waveform.

   1st Harmonic : Smoothest

   2nd " : Sharper

   5th " : Sharpest

3. If all harmonics are removed except the first one then the curve will be a perfect and smooth ~~sin~~ cosine curve.

Complex Exponential FS

Fourier series is used to represent periodic signal as a sum of complex exponential.

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

where c_k = Fourier coefficient

$$= \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt$$

C_K → How much the complex exponential at frequ exists in the signal

K → can be positive or negative

Parseval's Relation: The total average power of signal is equal to the sum of the powers of all harmonic components.

$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{K=-\infty}^{\infty} |C_K|^2$$

Math (Parseval's relation):-

① $x(t) = A\cos(\omega_0 t)$ → verify Parseval's relation.

From the parseval's theorem we get:-

$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{K=-\infty}^{\infty} |C_K|^2$$

$$L.H.S = \frac{1}{T}\int_T |x(t)|^2 dt = \frac{1}{T}\int A^2\cos^2\omega_0 t\, dt$$

$$= \frac{A^2}{T}\int_0^T \frac{1+\cos 2\omega_0 t}{2} dt$$

$$= \frac{A^2}{2T}\int_0^T dt + \frac{A^2}{2T}\int_0^T \cos 2\omega_0 t\, dt$$

$$= \frac{A^2}{2T}[t]_0^T + 0 = \frac{A^2}{2T} \times T = \frac{A^2}{2}$$

$$x(t) = A\cos\omega_0 t$$

$$= A\left(\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}\right)$$

$$= \frac{A}{2}e^{j\omega_0 t} + \frac{A}{2}e^{-j\omega_0 t}$$

Comparing $\sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$ with $\frac{A}{2} e^{j\omega_0 t}$

where $c_k = \frac{A}{2}$ $k = 1$

$c_1 = \frac{A}{2}$

Again, Comparing $\sum_{k=-\infty}^{\infty} c_k e^{-jk\omega_0 t}$ with $\frac{A}{2} e^{-j\omega_0 t}$

where $c_k = \frac{A}{2}$ $k = -1$, $c_1 = A/2$

$$R.H.S = \sum_{k=-\infty}^{\infty} |c_k|^2 = |c_1|^2 + |c_{-1}|^2 = \frac{A^2}{4} + \frac{A^2}{4} = \frac{A^2}{2}$$

L.H.S = R.H.S so Parseval's theorem satisfied

(ii) $x(t) = A \sin(\omega_0 t)$ → verify parseval's relation:-

From the perseval's relation we get:-

$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |c_k|^2$$

$$L.H.S = \frac{1}{T}\int_T A^2 \sin^2 \omega_0 t \, dt = \frac{A^2}{T}\int_0^T \frac{1-\cos 2\omega_0 t}{2} dt$$

$$= \frac{A^2}{2T}\int_0^T dt - \frac{A^2}{2T}\int_0^T \cos 2\omega_0 t \, dt$$

$$= \frac{A^2}{T}$$

$$x(t) = A \sin(\omega_0 t)$$

$$= A\left\{\frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}\right\}$$

$$= \frac{A}{2j} e^{j\omega_0 t} - \frac{A}{2j} e^{-j\omega_0 t}$$

$\sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$ comparing with $\frac{A}{2j} e^{j\omega_0 t}$

where $k=1$; $c_1 = \frac{A}{2j}$

$\sum_{k=-\infty}^{\infty} c_k e^{-jk\omega_0 t}$ comparing with $\frac{-A}{2j} e^{-j\omega_0 t}$

$k=-1$ $c_{-1} = \frac{-A}{2j}$

$$R.H.S = \sum_{k=-\infty}^{\infty} |c_k|^2 = |c_1|^2 + |c_{-1}|^2 = \left|\frac{A^2}{4j^2}\right| + \left|\frac{+A^2}{4j^2}\right|$$

$$= \left|\frac{A^2}{4}\right| + \left|-\frac{A^2}{4}\right| = \frac{A^2}{2}$$

As L.H.S = R.H.S Parseval theorems are satisfied

Symmetry properties in F.S:-

Even signal $x(t) = x(-t)$; $b_n = 0$

odd signal $x(t) = -x(-t)$; $a_n = 0$, $a_0 = 0$

Proof even signal $b_n = 0$

$x(t) = \cos t$

$b_n = \frac{2}{T}\int_0^T x(t) \sin(n\omega_0 t)\, dt$

$= \frac{2}{T}\int_0^T \underbrace{\cos t}_{\text{even}} \underbrace{\sin(n\omega_0 t)}_{\text{odd}}\, dt$

$= \frac{2}{T} \times 0 = 0$

even × even odd = odd

odd function का integral 0

Proof odd signal $a_n = 0$

$x(t) = \sin t$

$a_n = \frac{2}{T}\int_0^T x(t) \cos(n\omega_0 t)\, dt$

$= \frac{2}{T}\int_0^T \underbrace{\sin t}_{\text{odd}} \underbrace{\cos(n\omega_0 t)}_{\text{even}}\, dt$

$= \frac{2}{T} \times 0$

$= 0$

$a_0 = \frac{1}{T}\int_0^T x(t)\, dt$

$= \frac{1}{T}\int_0^T \sin t\, dt$

$= \frac{1}{T} \times 0 = 0$

Fourier Transform

Fourier Series is for periodic signal

Fourier Transform is for aperiodic signal

Fourier transform $X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$

Inverse F.T $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$

Linear Property

If $x(t) \longleftrightarrow X(\omega)$

$y(t) \longleftrightarrow Y(\omega)$

$a x(t) + b y(t) \longleftrightarrow a X(\omega) + b Y(\omega)$

Time scaling

$x(t) \longleftrightarrow X(\omega)$

$x(at) \longleftrightarrow \frac{1}{|a|} X(\omega/a)$

Time shifting:

If $x(t) \longleftrightarrow X(\omega)$

$x(t-t_0) \longleftrightarrow e^{-j\omega t_0} X(\omega)$

Time Reversal

If $x(t) \longleftrightarrow X(\omega)$

$x(-t) \longleftrightarrow X(-\omega)$

Existence Condition $\int_{-\infty}^{\infty} |x(t)|\, dt < \infty$

\# $x(t) = \delta(t)$

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t)\, e^{-j\omega t}\, dt = \int_{-\infty}^{\infty} \left[e^{-j\omega t}\right]_{t=0} = 1$$

\# $x(t) = e^{-at}\, u(t)$

$$X(\omega) = \int_{0}^{\infty} e^{-at}\, e^{-j\omega t}\, dt$$

$$= \int_{0}^{\infty} e^{-(a+j\omega)t}\, dt$$

$$= \frac{1}{-(a+j\omega)} \left[e^{-(j\omega+a)t}\right]_{0}^{\infty}$$

$$= \frac{1}{a+j\omega}$$ Ans

1. DTFT: Discrete Time Infinite sequence
2. DFT: for finite length sequence

DTFT	DFT
1. Used for infinite-length sequence	1. Used for finite length sequence..
2. Frequency is continuous	2. Frequency is discrete
3. Output is a continuous curve	3. Output is finite points
4. Mostly theoretical analysis	4. Mostly practical/Computer implementation.
5. Not directly computable on computers	5. Easily computable using FFT

(Discrete ~~Time~~ Fourier Transform)

DFT

$$X[K] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N}Kn} \; ; \; K = 0, 1, \ldots, N-1$$

Inverse DFT:-

$$x[n] = \frac{1}{N} \sum_{K=0}^{N-1} X[K]\, e^{j\frac{2\pi}{N}Kn} \; ; \; n = 0, 1, \ldots, N-1$$

V.V.I Find the DFT of $x[n] = \{1, 1, 1, 1\}$

$$X[K] = \sum_{n=0}^{N-1} x[n]\, e^{-j\omega Kn}$$

$$= \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N}Kn}$$

$$X[K] = \sum_{n=0}^{3} x[n]\, e^{-j\frac{2\pi}{4_2}Kn}$$

$$= \sum_{n=0}^{3} x[n]\, e^{-j\frac{\pi}{2}Kn}$$

for $K=0$;

$n=0$; (1) $e^{-j\frac{\pi}{2}\times 0\times 0} = 1 \cdot e^{0} = 1$

$n=1$; (1) $e^{-j\pi/2\times 0\times 1} = 1 \cdot e^{0} = 1$

$n=2$; (1) $e^{-j\pi/2\times 0\times 2} = 1 \cdot e^{0} = 1$ → SUM = 4

$n=3$; (1) $e^{-j\pi/2\times 0\times 3} = 1 \cdot e^{0} = 1$ $X[0] = 4$

for $K=1$;

$n=0$; (1) $e^{-j\frac{\pi}{2}\times 1\times 0} = (1) \cdot e^{0} = 1$

$n=1$; (1) $e^{-j\frac{\pi}{2}\times 1\times 1} = e^{-\frac{j\pi}{2}}$

$n=2$; (1) $e^{-j\frac{\pi}{2}\times 1\times 2} = e^{-j\pi}$

$n=3$; (1) $e^{-j\frac{\pi}{2}\times 1\times 3} = e^{\frac{-j\pi 3}{2}}$

$$X[1] = 1 + e^{-j\pi/2} + e^{-j\pi} + e^{-\frac{j3\pi}{2}}$$

$$= 1 + (-j) + (-1) + j = 0$$

$e^{-j\pi/2} = \cos(\pi/2) - j\sin(\pi/2)$

$= 0 - j(1) = -j$

for $K=2$;

$n=0$; (1) $e^{-j\frac{\pi}{2}\times 2\times 0} = (1).e^{0} = 1$

$n=1$; (1) $e^{-j\frac{\pi}{2}\times 2\times 1} = e^{-j\pi}$

$n=2$; (1) $e^{-j\frac{\pi}{2}\times 2\times 2} = e^{-j2\pi}$

$n=3$; (1) $e^{-j\frac{\pi}{2}\times 2\times 3} = e^{-j3\pi}$

$X[2] = 1 + e^{-j\pi} + e^{-j2\pi} + e^{-j3\pi}$

$= 1 + (-1) + 1 - 1 = 0$

for $K=3$;

$n=0$; (1) $e^{-j\pi/2\times 3\times 0} = e^{0} = 1$

$n=1$; (1) $e^{-j\frac{\pi}{2}\times 3\times 1} = e^{-j\frac{3\pi}{2}}$

$n=2$; (1) $e^{-j\frac{\pi}{2}\times 2\times 3} = e^{-j3\pi}$

$n=3$; (1) $e^{-j\frac{\pi}{2}\times 3\times 3} = e^{-j\frac{9\pi}{2}}$

$X[3] = 1 + e^{-j\frac{3\pi}{2}} + e^{-j3\pi} + e^{-j\frac{9\pi}{2}}$

$= 1 + j - 1 - j = 0$

$X[K] = \{4, 0, 0, 0\}$

Problem Find the DFT of $\{1, -1, 1, -1\}$

$$X[K] = \sum_{n=0}^{3} x[n]\, e^{-j\frac{\pi}{2}Kn}$$

for $K=0$;

$n=0$; (1) $e^{-j\frac{\pi}{2}\times 0\times 0} = 1$ $\quad X[0]=0$

$n=1$; (-1) $e^{j\pi/2\times 0\times 1} = -1$

$n=2$; (1) $e^{-j\pi/2\times 0\times 2} = 0+1=1$

$n=3$; (-1) $e^{-j\pi/2\times 0\times 3} = -1$

for $K=1$;

$n=0$; (1) $e^{-j\pi/2 \times 0} = 1$

$n=1$; (-1) $e^{-j\frac{\pi}{2} \times 1 \times 1} = -e^{-j\pi/2}$

$n=2$; (1) $e^{-j\pi/2 \times 2 \times 1} = e^{-j\pi}$

$n=3$; (-1) $e^{-j\pi/2 \times 3 \times 1} = -e^{-3j\pi}$

$$X[1] = 1 + (-e^{-j\pi/2}) + e^{-j\pi} + (-e^{-3j\pi})$$

$$= 1 + (+j) + (-1) + (-j)$$

$$= 0$$

for $K=2$;

$n=0$; (1) $e^{-j\frac{\pi}{2} \times 2 \times 0} = 1$

$n=1$; (-1) $e^{-j\frac{\pi}{2} \times 2 \times 1} = -e^{-j\pi}$

$n=2$; (1) $e^{-j\frac{\pi}{2} \times 2 \times 2} = e^{-j2\pi}$

$n=3$; (-1) $e^{-j\frac{\pi}{2} \times 2 \times 3} = -e^{-j3\pi}$

$$X[2] = 1 - e^{-j\pi} + e^{-j2\pi} - e^{-j3\pi}$$

$$= 1 + 1 + 1 + 1 = 4$$

for $K=3$;

$n=0$; (1) $e^{-j\frac{\pi}{2} \times 0 \times 3} = 1$

$n=1$; (-1) $e^{-j\frac{\pi}{2} \times 3 \times 1} = -e^{-j\frac{3\pi}{2}}$

$n=2$; (1) $e^{-j\frac{\pi}{2} \times 2 \times 3} = +e^{-j3\pi}$

$n=3$; (-1) $e^{-j\frac{\pi}{2} \times 9} = -e^{-j\frac{9\pi}{2}}$

$$X[3] = 1 - e^{-j\frac{3\pi}{2}} + e^{-j3\pi} - e^{-j\frac{9\pi}{2}}$$

$$= 1 - j - 1 + j = 0$$

$$X[K] = \{0, 0, 4, 0\}$$

Multiplication of Two DFTs and circular convolution

Suppose that we have two finite duration sequences of length N, $x_1(n)$, $x_2(n)$

$$X_1[K] = \sum_{n=0}^{N-1} x_1(n)\, e^{-j 2\pi n \frac{K}{N}}$$

$$X_2[K] = \sum_{n=0}^{N-1} x_2(n)\, e^{-j 2\pi n \frac{K}{N}}$$

$$X_3(K) = X_1(K) \cdot X_2(K)$$

$$x_3(m) = \frac{1}{N} \sum_{K=0}^{N-1} X_3[K]\, e^{j 2\pi K \frac{m}{N}}$$

$$= \frac{1}{N} \sum_{K=0}^{N-1} X_1[K]\, X_2[K]\, e^{j 2\pi K \frac{m}{N}}$$

$$= \frac{1}{N} \sum_{K=0}^{N-1} \left[\sum_{n=0}^{N-1} x_1(n)\, e^{-j 2\pi n \frac{K}{N}}\right]\left[\sum_{n=0}^{N-1} x_2(n)\, e^{-j 2\pi m \frac{K}{N}}\right]$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \sum_{l=0}^{N-1} x_2(l) \left[\sum_{K=0}^{N-1} e^{j 2\pi K (m-n-l)/N}\right]$$

$$\sum_{K=0}^{N-1} a^K = \begin{cases} N; & a = 1 \\ \frac{1-aN}{1-a}; & a \neq 1 \end{cases}$$

Let, $a = e^{j 2\pi (m-n-l)/N}$

Since, m, n, l are integers.

$$\frac{m-n-l}{N} = \pi$$

$$e^{j 2\pi \pi} = \cos(2\pi\pi) + j \sin(2\pi\pi)$$

$$= 1 + 0 = 1$$

When $a = 1$;

$$a = e^{j 2\pi (m-n-l)/N}$$

To become $a=1$,

$\{(m-n-l)/N\}$, integers

$$\frac{m-n-l}{N} = p$$

$$m-n-l = pN$$

$$l = m-n-pN$$

$$l = (m-n) \pmod N$$

$$\boxed{l = ((m-n))_N}$$

$$\sum_{k=0}^{N-1} a^k = \begin{cases} N; & l = ((m-n))_N \\ 0, & \text{otherwise} \end{cases}$$

$$\boxed{x_3(m) = \sum_{n=0}^{N-1} x_1(n)\, x_2((m-n))_N}$$

Circular Convolution.

Ex: 5.2.1 Perform the circular convolution of the

$x_1(n) = \{2, 1, 2, 1\}$

$x_2(n) = \{1, 2, 3, 4\}$

$$x_3(m) = \sum_{n=0}^{N-1} x_1(n)\, x_2((m-n))_N$$

$m=0$; $n=0$: $x_1(0)\, x_2((0))_4 = x_1(0)\, x_2(0) = 2 \times 1 = 2$

$n=1$: $x_1(1)\, x_2((-1))_4 = x_1(1)\, x_2(3) = 1 \times 4 = 4$

$n=2$: $x_1(2)\, x_2((-2))_4 = x_1(2)\, x_2(2) = 2 \times 3 = 6$

$n=3$; $x_1(3)\, x_2((-3))_4 = x_1(3)\, x_2(1) = 1 \times 2 = 2$

$x_3[0] = 14$; $y[0] = 14$

$m=1$; $n=0$; $x_1(0)\, x_2((1))_4 = x_1(0)\, x_2(5) = 2 \times 0 = 4$

$n=1$; $x_1(1)\, x_2((0))_4 = x_1(1)\, x_2(4) = 1 \times 4 = 4$

$n=2$; $x_1(2)\, x_2((-1))_4 = x_1(2)\, x_2(3) = 2 \times 4 = 8$

$n=3$; $x_1(3)\, x_2((-2))_4 = x_1(3)\, x_2(2) = 1 \times 3 = 3$

$y[1] = 16$

m=2;

n=0; $x_3(0)\ x_2((2))_4 = x_1(0)\ x_2(2) = 2 \times 3 = 6$

n=1; $x_1(0)\ x_2((1))_4 = x_1(1)\ x_2(1) = 1 * 2 = 2$

n=2; $x_1(2)\ x_2((0))_4 = x_1(2)\ x_2(0) = 2 \times 1 = 2$

n=3; $x_1(3)\ x_2((-1))_4 = x_1(3)\ x_2(3) = 1 \times 4 = 4$

$y[2] = 14$

m=3;

n=0; $x_1(0)\ x_2((3))_4 = x_1(0)\ x_2(3) = 2 \times 4 = 8$

n=1; $x_1(1)\ x_2((2))_4 = x_1(1)\ x_2(2) = 1 \times 3 = 3$

n=2; $x_1(2)\ x_2((1))_4 = x_1(2)\ x_2(1) = 2 \times 2 = 4$

n=3; $x_1(3)\ x_2((0))_4 = x_1(3)\ x_2(0) = 1 \times 1 = 1$

$y[3] = 16$

$y[n] = \{14, 16, 14, 16\}$ Ans

Discrete Time Fourier Transform (DTFT):-

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n}$$

$X(\omega)$ is a complex function of real frequency variable and can be written as:-

$$X(\omega) = X_{re}(\omega) + j\, X_{img}(\omega)$$

where $X_{re}(\omega)$, $X_{img}(\omega)$ are real and imaginary parts of $X(\omega)$ respectively.

$$X_{re}(\omega) = |X(\omega)| \cos\theta(\omega)$$

$$X_{img}(\omega) = |X(\omega)| \sin\theta(\omega)$$

$$|X(\omega)|^2 = |X_{re}(\omega)|^2 + |X_{im}(\omega)|^2$$